MILITARY SPECIFICATION

TRANSMITTER, RATE OF FLOW

This specification is approved for use within the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1. Scope. This specification covers the requirements for turbo-jet engine, aircraft rate-of-fuel flow mass transmitters, designated TRU-31A/A.

1.2 Classification. Transmitters shall be one of the following types, as specified (see 6.2):

TRU-31A/A-1	0 to 1,200 pph range
TRU-31A/A-2	0 to 1,800 pph range
TRU-31A/A-3	0 to 2,500 pph range
TRU-31A/A-4	0 to 800 pph range
TRU-31A/A-5	0 to 4,000 pph range
TRU-31A/A-6	0 to 5,000 pph range
TRU-31A/A-7	0 to 400 pph range
TRU-31A/A-8	0 to 5,000 pph range

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Federal

PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-636	Box, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Logistics Center/MMEDO, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letters.

AMSC N/A FSC 6620

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Military

MIL-P-116	Preservation, Methods Of
MIL-C-5541	Chemical Films and Chemical Film Materials For Aluminum and Aluminum Alloys
MIL-T-5624	Turbine Fuel, Aviation Grades JP-4 and JP-5
MIL-A-8625	Anodic Coatings, For Aluminum and Aluminum Alloys
MIL-I-27304	
MIL-B-27497	
MIL-M-45202	

STANDARDS

Military

DOD-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Markings of U.S. Military Property
MIL-STD-143	Standards And Specifications, Order of Precedence For The Selection Of
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements For Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-470	Maintainability Program Requirements (For Systems and Equipment)
MIL-STD-471	Maintability Demonstration
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-810	Environmental Test Methods
MIL-STD-889	Dissimilar Metals
MS24266	Connectors, Plug, Electrical-Straight, Miniature
MS27034	Connectors, Receptacle, Electrical, Pin Insert, Cylindrical, Miniature Hermetic, Solder Mount
MS33514	Fitting End, Standard Dimensions For Flareless Tube Connections and Gasket Seal
MS33649	Bosses, Fluid Connection-Internal Straight Thread
MS33656	Fitting End, Standard Dimensions For Flared Tube Connection and Gasket Seal

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(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

ASTM D3951 Packaging, Commercial

(Application for copies should be addressed to: ASTM, 1916 Race St, Philadelphia PA 19103).

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be availble in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede application laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first artice inspection (see 4.4 and 6.3).

3.2 Selection of standards and specifications. Standards and specifications for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials

3.3:1 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the transmitters except where magnetic materials are essential.

3.3.2 Nonferrous materials. Nonferrous materials shall be used for all parts of the transmitters except where ferrous materials are essential.

3.3.3 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service.

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3.3.4 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.3.5 Toxic and corrosive fumes. The transmitter shall be such material and construction that it will not produce corrosive or deleterious fumes, acid, or alkali under normal use or under the conditions specified herein.

3.3.6 Protective treatment. When materials are used in the construction of the transmitter that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.3.7 Titanium. Where high temperature and other strigent environmental conditions are a factor in the design of the transmitter, the use of titanium will be permitted, with the approval of the contracting activity where lightweight and highstrength materials are required. Use of lightweight materials and weight-saving designs shall be of major consideration and their use shall be investigated and exploited to the greatest possible extent.

3.3.8 Recycled and reclaimed materials. The use of recycled and reclaimed materials shall be encouraged to the maximum extent possible.

3.4 Design and construction. The transmitters shall be designed to operate satisfactorily with fuels having any of the variation in characteristics specified in MIL-T-5624. The fuel flow transmitter shall be calibrated for rate of flow in pounds per hour of aviation fuel in accordance with MIL-T-5624. The specific gravity of the fuel shall be 0.785 $\pm$ 0.005 at 15.6°C. An approved commercial fluid of higher flash poinit may be used for the testing purposes. The calibration of the transmitter shall be such that it will satisfactorily meet the requirements specified herein when tested with fuels in accordance with Mil-T-5624. The transmitter shall measure the fuel flow rate on a gravimetric basis without separate and distinct correcting means.

3.4.1 Reliability. The transmitter shall have a minimum acceptable mean-time-between-failures (MTBF) of 1,200 hours during its service life demonstrated to a 90 percent confidence level.

3.4.2 Longevity. The transmitter shall have an operating life span (equipment longevity as defined in MIL-STD-781) of not less than 2,500 hours before wearout failures occur or the equipment consistently fails to meet the specified MTBF index. Parts requiring replacement within this period of time and the normal operating period of such parts shall be reported by the contractor to the contracting activity.

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3.4.3 Maintainability. Maintainability shall be in accordance with MIL-STD-470.

3.4.4 System compatibility. The TRU-31A/A-1 through -3 transmitters shall be interchangeable with previously manufactured TRU-31/A-1 through -3 transmitters and shall be completely compatible with the respective EFU-15/A-1 through -9 and EFU-15A/A-1 through -9 indicators conforming to MIL-I-27304. The TRU-31A/A-4, -5 and -6 and transmitters shall be interchangeable with previously manufactured TRU-31A/A-4, -5, and -6 transmitters and shall be completely compatible with the EFU-15A/A-10 through -18 indicators. The TRU-31A/A-7 transmitter shall be compatible with the EFU-15A/A-19 through -21 indicators.

3.4.5 Power. The transmitter shall be designed to operate from 115V ±20 percent 400-Hz ±20 percent single-phase a-c power having characteristics in accordance with MIL-STD-704. The transmitter shall meet the performance criteria specified herein when the input power is varied at any combination of voltages and frequencies ranging from 104 to 126V and 320 to 480 Hz with frequency spikes appearing from 300 to 500 Hz and voltage spikes appearing up to 200V. The transmitter shall not be damaged by the application of voltages and frequencies of 10 percent more or less than the specified operating range.

3.4.6 Operation. The transmitter shall operate on the second harmonic principle in which the moving element consists of a nonenergized permanent 2-pole magnet (or magnets) and the toroidally wound stator is energized with 115V 400-Hz single-phase power. An externally located power supply shall provide a 75V ±20 percent, 6 ±0.5 percent cycle 2-phase output to drive the transmitter impeller motor at a constant speed. The power supply in the compatible indictor shall provide electrical power to drive the transmitter impeller motor at a constant speed.

3.4.6.1 The transmitter output signal shall be capable of operating two indicators simultaneously within 1/2° of single indicator signal effect as a result of this dual operation.

3.4.6.2 For an increasing flow rate, the direction of rotation of the transmitter impeller rotor shall be clockwise as viewed from the inlet end. The pickoff rotor shall rotate 307°.

3.4.6.3 A rugged mechancial stop shall be provided with the transmitter above the maximum flow rate value but below the zero point. The transmitter shall not be damaged by fuel-flow rates exceeding 20 percent of the rated value provided the pressure rating of the transmitters is not exceeded.

3.4.6.4 Operating pressure. The transmitters shall be designed for operation in fuel systems with a normal fuel pressure up to 1,000 psi. It shall be capable of withstanding, without operational damage, a static fuel pressure of 1,600 psi at both room temperature and -54°C, and a static fuel pressure of 1,000 psi at a fuel temperature of +71°C when in an ambient temperature of 140°C. The transmitter shall also be capable of withstanding a static fuel pressure of 3,000 psi (proof test) at room temperature.

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3.4.7 Pressure drop. The transmitter shall be so designed that when inoperative under any fuel-flow conditions, the pressure drop will not be greater than when the transmitter is operating normally. Table I specifies the allowable pressure drop.

TABLE I. Pressure Drop.

Flow PPH	Max Pressure Drop Psi
400	0.20
800	0.20
1,200	0.40
1,800	0.85
2,500	1.50
4,000	4.25
5,000	6.75

3.4.8 Internal wiring. Internal wiring for all TRU-31A/A transmitters shall be identical. The wires shall be color coded and connected as indicated in table II. The system electrical connections for the TRU-31A/A-1,-2,-3,-4,-7 and -8 transmitters are shown on figure 1. The system electrical connections for the TRU-31A/A-5 and -6 transmitters are shown on figure 2.

TABLE II. TRU-31A/A Transmitter Connections.

Pin		FUNCTION	Color Coding
Type 1-4,-7 and -8	Type 5-6		
1	A	Input power to pickoff 115V 400 Hz ground	Green
2	B	Input to pickoff 115 400Hz positive	Red
3	C	Output signal from pickoff, second harmonic	Blue
4	D	Output signal from pickoff, second harmonic	White
5	E	Input power to motor 2 phase 6 Hz 65V	Gray
6	F	Input power to motor 2 phase 6 Hz 65V	Black
7		Internal ground	

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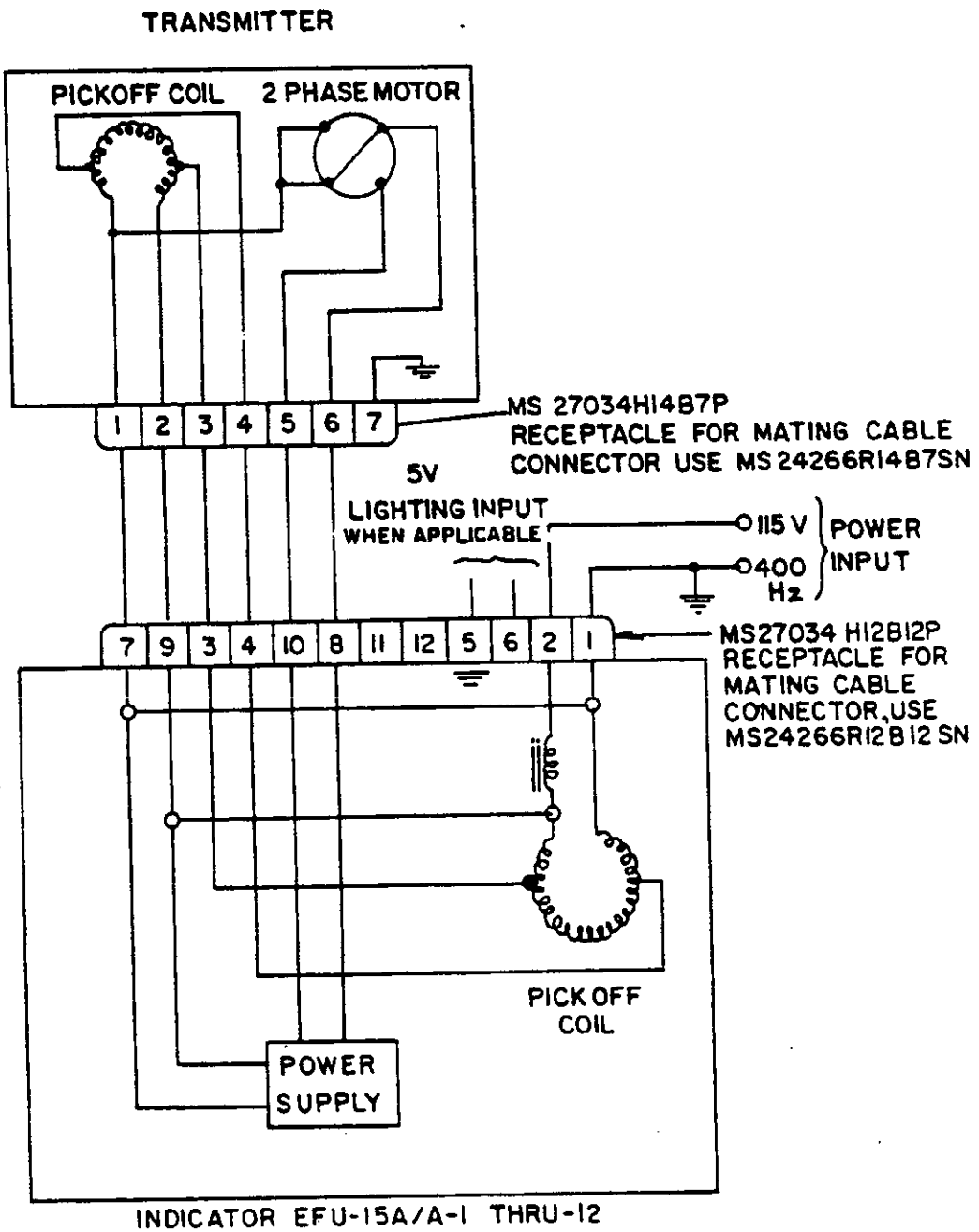


FIGURE 1. System connection diagram for TRU -31A/A-1,-2,-3,-4-7 and -8.

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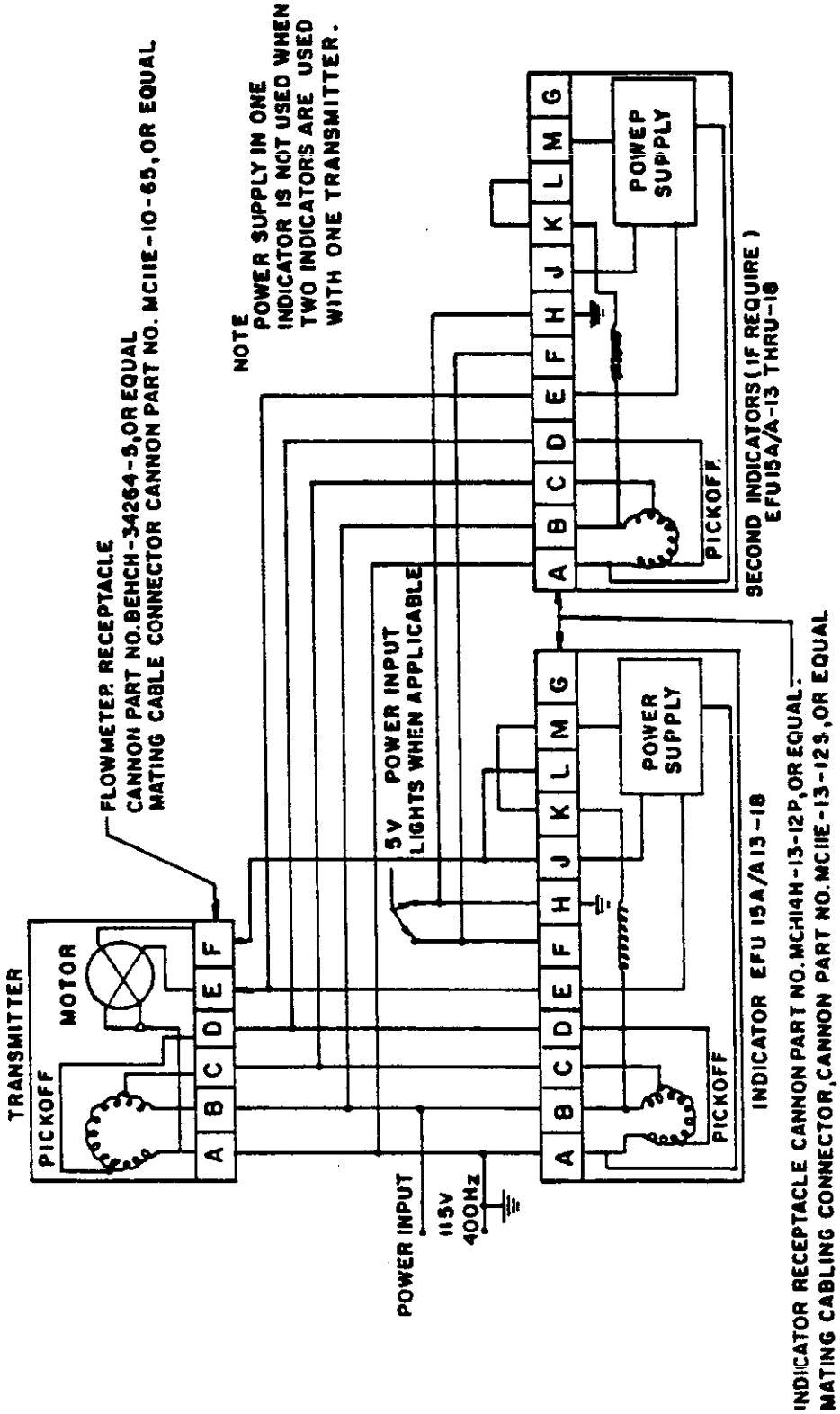


FIGURE 2. System connection diagram for TRU-31A/A-5 and -6 Transmitters.

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3.4.9 Electrical receptacles. The transmitter shall be provided with a hermetically sealed miniature electrical receptacle in accordance with MIl-C-26500 which shall be designed to meet the environmental requirements specified herein. TRU-31A/A-1, -2, -3, -4, -7 and -8 transmitters shall utilize an MS27034H14B7P electrical connector, and the TRU-31A/A-5 and -6 transmitters shall utilize Cannon electrical connector BFMCH-34264-5 or an approved equivalent.

3.5 Performance

3.5.1 Tolerance. When subjected to the tests specified herein, the flow rates for the tansmitter specified shall be within the tolerances shown in the applicable column of Tables I, III, and IV.

3.5.2 Vibration. The transmitter shall operate normally at 20g in the frequency range from 5 to 2,000 Hz within the limits of this specification.

3.5.3 Power consumption. The power consumption of the transmitter pickoff coil shall not exceed 14 va with rated power applied.

3.5.4 Environmental conditions. The transmitter shall meet the requirements specified herein under the following conditions:

a. Altitude-Altitudes ranging from sea level to 100,000 feet

b. Shock-Shock forces up to 25g

c. Salt fog-Salt fog for a period of 50 hours

d. Humidity-Relative humidity up to 95 percent

e. Fungus-Fungus growth as encountered in tropical climates.

3.5.5 Dielectric strength. The transmitter shall withstand 500V at a commercial frequency applied between any pin on the electrical receptacle and the case for 5 seconds.

3.5.6 Electromagnetic interference. The transmitter shall meet the electromagnetic interference requirement of MIL-STD-461 for class ID equipment except the frequency range shall be from 0.15 to 150 MHz.

3.5.7 Response. The time required for the transmitter output signal to come within 1 percent of the final steady state shall not exceed 3 seconds under the conditions specified in 4.6.6.

3.5.8 Temperature range. The transmitter shall operate satisfactorily in ambient tempertures ranging from -54° to $+140^{\circ}$C with fuel through the transmitter. The operating fuel temperatures range shall be from -54° to $+71^{\circ}$C. The transmitter shall withstand exposure to a temperature of 190°C.

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TABLE III. Scale error tolerances +PPH.

Fuel Flow Test Points	400 PPH Range		800 PPH Range		1,200 PPH Range		1,800 PPH Range		2,500 PPH Range		4,000 PPH Range		5,000 PPH Range		5,000 PPH Range (-8 only)	
	RM	L&H	RM	L&H	RM	L&H	RM	L&H	RM	L&H	RM	L&H	RM	L&H	RM	L&H
0			10	15	15	20	22	25	30	32	35	40	40	45	40	50
50	5	6														
100	5	6	8	10												
150	3	4														
200	4	5	7	9	12	15	16	20	22	25	25	30	30	35	24	35
250	5	6	6	8												
300	5	6	5	7	8	13	15	20								
350	6	8	5	7												
400	7	8	5	7	8	13	14	18	20	25						
450			6	8												
500			7	9	7	10	14	18	17	25	22	30	30	35	16	25
550			8	10												
600			8	10	8	10	22	25	25	30	30	35	24	35		
650			10	12												
700			12	14	8	13	12	16					25	30	12	20
800			14	18	8	13	12	16	15	20						
900					10	15	14	18					25	30		
1,000					12	17	14	18	15	20	20	25	25	30	12	20
1,100					15	20										
1,200					25	30	20	25	17	25	18	25	25	30	12	20
1,400							25	30	22	30	18	25	25	35	12	24
1,600							30	35	26	35	20	25	25	40	16	28
1,800							35	45	30	40	25	30	30	40	19	28
2,000									32	45	30	40	35	50	22	35
2,200									35	50	30	45				
2,400									40	55	35	50				
2,500									50	60	40	55	45	60	27	40
2,600																
2,800											50	60	70	80	40	50
3,000																

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TABLE III. Scale error tolerances +pph (Cont).

Fuel Flow Test Points	400 PPH Range		800 PPH Range		1,200 PPH Range		1,800 PPH Range		2,500 PPH Range		4,000 PPH Range		5,000 PPH Range		5,000 PPH Range (-8 only)	
	RM	L&H	RM	L&H	RM	L&H	RM	L&H	RM	L&H	RM	L&H	RM	L&H	RM	L&H
3,500											60	70				
4,000											70	80	80	90	48	60
4,500													90	100	48	60
5,000																

TABLE IV. Test Tolerances.

Test	400	800	1,200	1,800	Tolerance +PPH 2,500	4,000	5,000
Voltage and frequency variation	2	2	4	6	8	10	12
Position error	2	2	4	5	7	8	8
Friction	2	2	4	5	6	7	8
Vibration							
Pointer oscillation	20	20	20	23	25	35	45
Pointer variation	20	20	20	23	25	35	45

3.6 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of DOD-STD-100 shall govern the manufactuer's part number and changes thereto.

3.7 Weight. The weight of the complete TRU 31A/A through -4 and -7 transmitters shall not exceed 1.5 pounds. The weight of the TRU-31A/A-5 and -6 transmitters shall not exceed 2.5 pounds. The weight of the TRU-31A/A-8 transmitter shall conform to figure 6.

3.8 Mounting. The transmitter end fittings shall be designed for connection to a 0.625 inch diameter fuel line. The flanges for the TRU-31A/A through -4 transmitters shall be in accordance with figure 3 and the TRU-31A/A-5 and -6 flanges shall be in accordance with figure 4.

3.9 Dimensions. The outline dimensions of the TRU-31A/A-1 through -4 and -7 transmitters shall conform to figure 3. The dimensions of the TRU-31A/A-5 and -6 transmitters shall conform to figure 6. The outline dimensions of the TRU-31A/A-8 transmitter shall conform to figure 6.

3.10 Synthetic jewel bearings. When synthetic sapphire or ruby jewel bearings are used, they shall be in accordance with MIL-B-27497.

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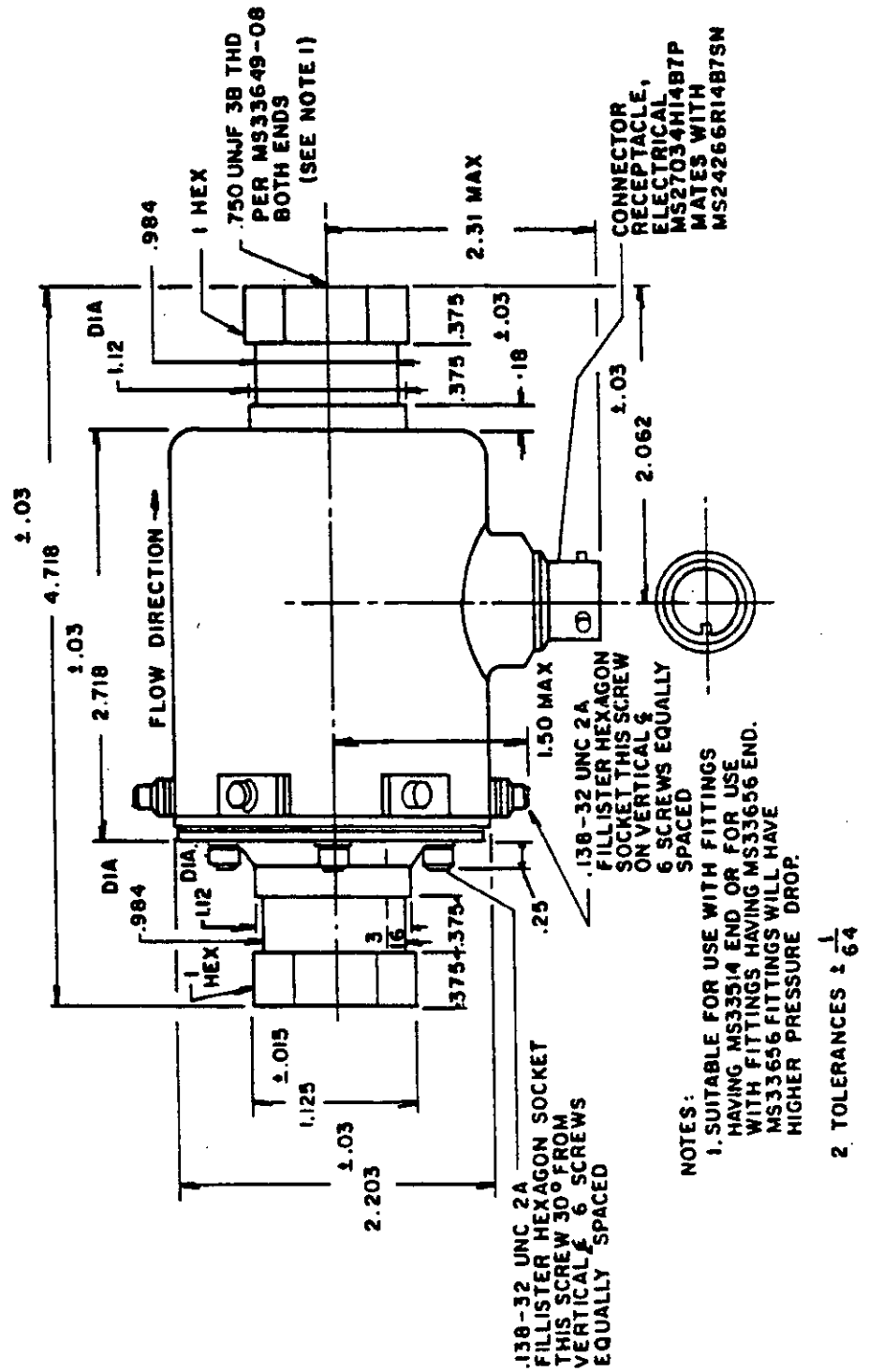


FIGURE 3. Transmitter outline - TRU-31A/A-1 Thru - 4 and -7.

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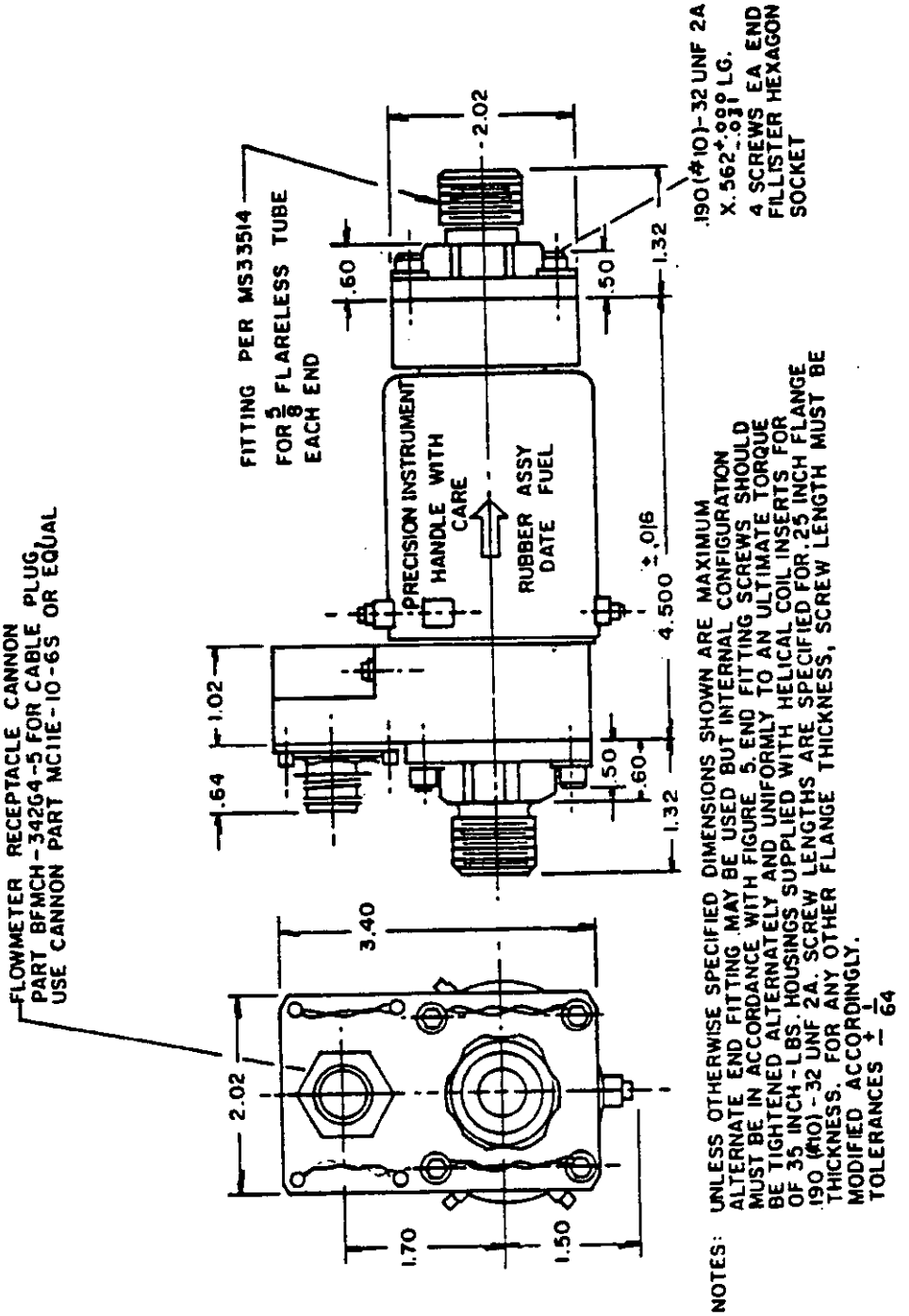
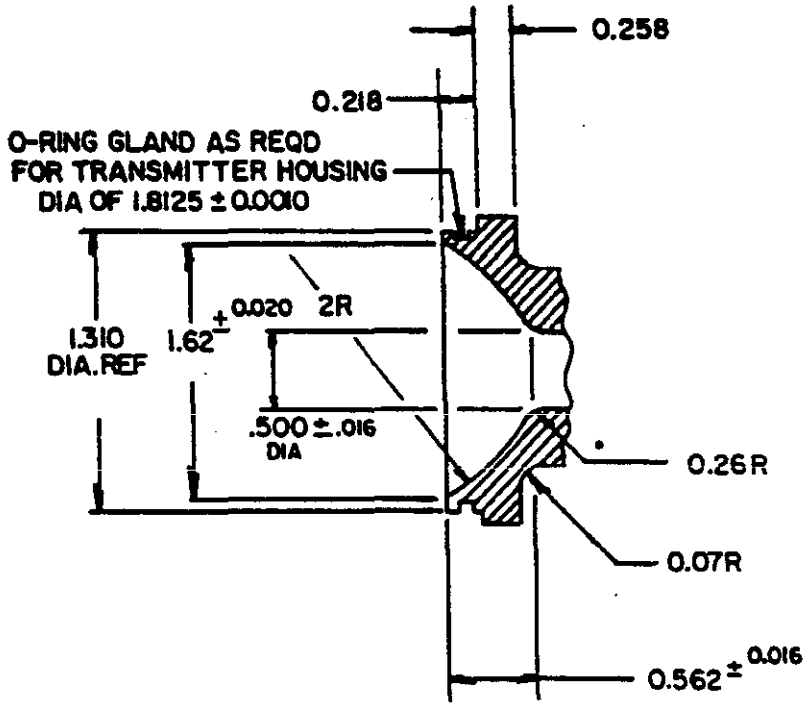


FIGURE 4. TRU-31A/A-5 and 6 Transmitter outline.

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SEE NOTE 2 OF FIGURE 4

FIGURE 5. End fitting detail.

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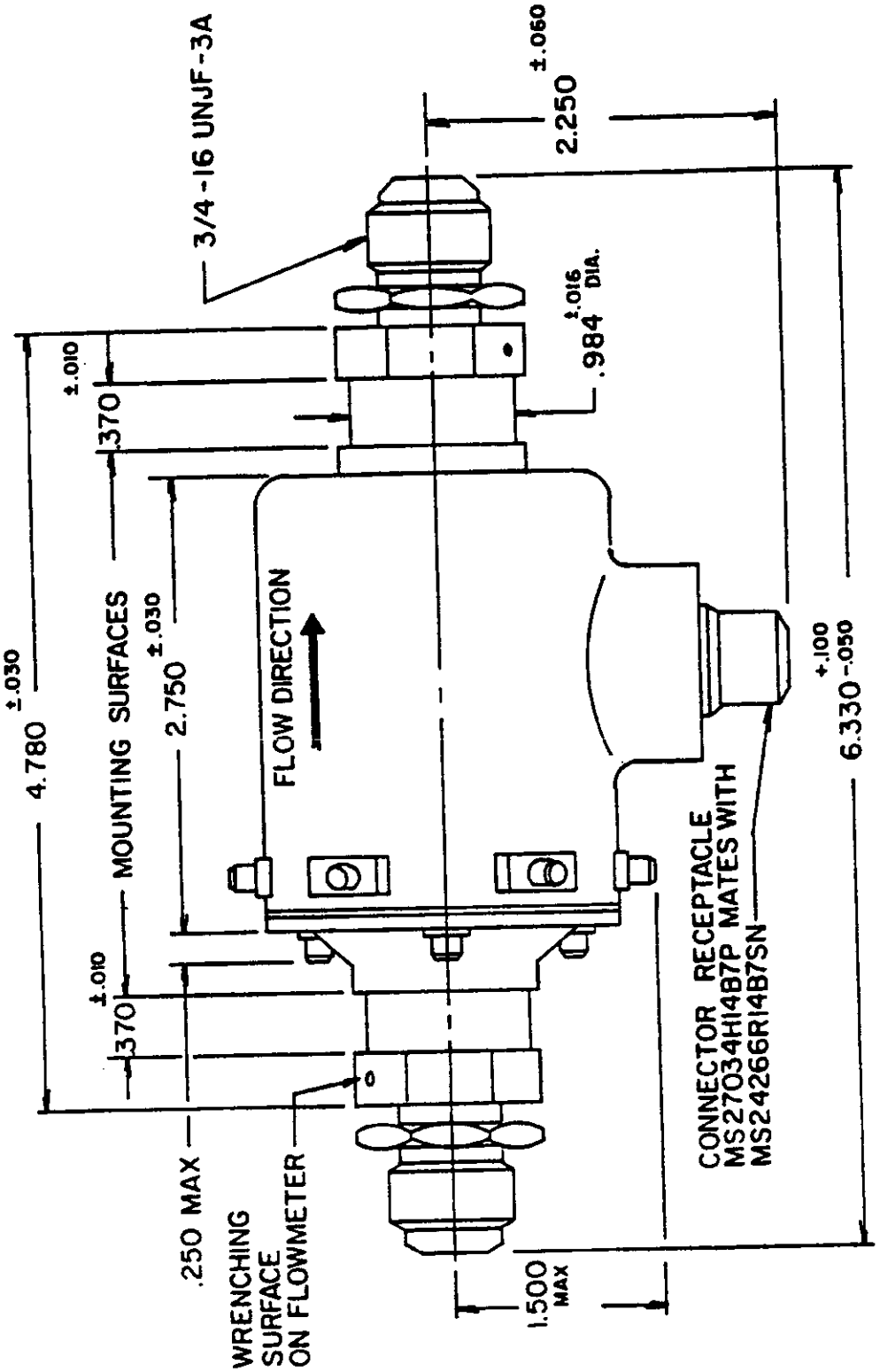


FIGURE 6. Transmitter outline - TRU-31A/A-8.

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3.11 Flow direction. The direction of flow through the transmitter shall be permanently and legibly marked by an arrow on both the front and back of the transmitters.

3.12 Caps. The fuel-line connections shall be provided with protective shipping caps or plugs to prevent foreign material from entering the transmitter.

3.13 Finishes and protective coating.

3.13.1 Alloy parts. Where practicable, aluminum or titanium alloy part shall be covered with an anodic film conforming to MIL-A-8625. Small holes, inserts, and small areas that must be machined after finishing need not be covered with a protective film.

3.13.2 Chemical treatment. Where the primary purpose of the treatment is to afford a suitable paint base, or for castings containing nonaluminum or titanium alloy inserts, a chemical treatment in accordance with MIL-C-5541 may be used in lieu of an anodic film for alloys that do not anodize satisfactorily.

3.13.3 Abrasion resistance. Where abrasion resistance is a factor, chemical film in accordance with MIL-C-5541 shall not be used.

3.13.4 Magnesium parts. Magnesium alloy parts shall be covered with an anodic film in accordance with MIL-M-45202, HAE process.

3.14 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIl-STD-130.

3.14.1 Nameplate. A nameplate shall be provided in accordance with MIl-STD-130. Metal decalcomanias shall not be used.

3.14.2 Schematic. An electric schematic showing the function of each connector pin shall be permanently attached to the body of the transmitter.

3.15 Shipment. The mounting flanges for the transmitter are considered an integral parts of the transmitter. The transmitter shall be fully assembled as shown on the applicable figure and shipped with protective shipping caps on both the inlet and outlet ports.

3.17 Workmanship. The transmitter, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner with MIL-STD-454, requirement 9. Particular attention shall be given to neatness and thoroughness of soldering, brazing, painting, riveting, machine-screw assemblies, and freedom of parts from burrs and sharp edges.

3.17.1 Dimensions and tolerances. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the transmitter, they shall be held or limited accordingly.

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4.1 Responsibility for inspection. Unless otherwise specified in the contract or order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsiblity for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contrator of the responsiblity of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of tests. The inspection and testing of transmitters shall be classified as follows:

a. First article inspection (see 4.4)

b. Quality conformance tests (see 4.5)

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in this specification.

4.3.1 Test power. The supply power shall be single-phase ac 115V, ±2 percent, and 400 Hz, controlled to less than ±1 Hz and of known value.

4.3.2 Test fuel. Unless otherwise specified, the fuel used for calibration in testing shall be JP-4 conforming to MIL-T-5624. The specific gravity of the fuel shall be 0.785 ±0.005 at 15.6°C. (Note: An approved commercial equivalent with higher flash point may be used.)

4.3.3 Test position. Unless otherwise specified, the transmitter shall be tested with the flow axis horizontal and the connector at the 3 o'clock position when viewed from the inlet end.

4.3.4 Test temperature. Unless otherwise specified, all tests shall be conducted at room temperature.

4.3.5 Test equipment. Unless otherwise specified, the transmitter shall be tested while connected to a master indicator and master power supply. The master power supply shall be so designed that it will have no frequency error, or if a frequency error exists, it shall be subtracted from the test date. All flow tests shall be performed on a primary flow stand which measured liquid mass flow rate directly in terms of weight and time.

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4.3.6 Test tolerances. Test tolerances shall be in accordance with Tables III and IV.

4.4 First articles inspection.

4.4.1 Test samples. The first article test samples shall consist of 14 transmitters representative of the production equipment. Eleven of the samples shall be subjected to the individual, reliability, and longevity tests only and the other three samples shall be subjected to all the tests except reliability and longevity. The samples shall be identified with the manufacturer's part number and such other information as required by the qualifying activity.

4.2.2 First article tests. The first article tests shall consit of all the tests specified uner 4.6. Unless otherwise specified, tests shall be conducted in the order determined by the qualifying activity.

4.5 Quality conformance tests. Quality conformance tests shall consist of:

a. Individual tests

b. Sampling tests

4.5.1 Individual tests. Each transmitter shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Dielectric strength

c. Scale error

d. Friction error

e. Leakage

f. Response time

g. Burn-in

4.5.2 Sampling tests. A minimum of 3 transmitters shall be selected at random from each 100 or fraction thereof produced on the contract and subjected to the following tests as described under 4.6:

a. Individual tests

b. Position error

c. Voltage and frequency variation

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c. Voltage and frequency variation

d. Voltage surge

e. Power consumption

f. Low temperature

g. High temperature

h. Vibration

i. Pressure drop

4.5.2.1 Reliability sampling. For reliability sampling, a lot shall consist of all transmitters produced on a specific contract. A minimum of 5 transmitters shall be selected at random from each 100 or fraction thereof produced on the contract and subjected to the test specified in 4.6.29.2.

4.5.2.2 Rejection and retest. When an item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, the tests specified by the contracting activity shall be repeated.

4.5.2.2.1 Reconditioning of tested transmitters. Unless otherwise specified, transmitters that have been subjected to the sampling tests shall be reconditioned by the contractor by replacing all worn or damaged parts. After reworking, the contractor shall resubmit the transmitters for acceptance.

4.5.2.3 Individual tests may continue. Individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or produced later shall not be made until it is determined that all items meet all requirements of the specification.

4.5.3 Defects in items already accepted. The invesigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the contracting activity of all defects likely to be found and the method of correcting them.

4.6 Test methods

4.6.1 Examination of product. The transmitter shall be inspected to determine compliance with the requirements specified herein with respect to materials, finish, workmanship, and marking. Each unit shall be weighed.

4.6.2 Dielectric strength. A potential of 500V at a commercial frequency shall be applied between any pin and the transmitter cae for a minimum of 5 seconds. There shall be no insulation breakdown as a result of this test.

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4.6.3 Scale error. The scale error of each transmitter shall be determined at the test points specified in Table III. The scale error shall be determined for both increasing and decreasing flow rates. Each test point in a set of readings shall be consistently approached from the same direction. A minimum of two sets of readings shall be taken at each test point. A set of readings shall consist of two flowstand weigh cycle times average to obtain average flow rates for two independent runs. The values for both runs however, shall be reported. If the individual readings for one of the runs exceeds or is within 5 percent of the specification limit, a second set of readings shall be taken. Scale error shall be determined while the transmitter is gently vibrated by an electromechanical vibrator mounted on the flow piping. The scale error test shall be conducted at room, low, and high temperatures, and substitute fuel operating conditions. Table III specifies the test tolerances.

4.6.4 Friction error. Friction error shall be determined at 25, 50, and 90 percent of full scale. The difference in readings taken before vibration and after vibration shall not exceed the values listed in Table IV. This test shall be combined with the scale error test.

4.6.5 Leakage. The transmitter shall be tested for leakage as follows: A nonshock hydraulic pressure shall be applied to the inlet connection with the outlet connection led free of air and capped. A pressure of 1,600 psi shall then be rapidly aplied 5 times at room temperature. After the last application of pressure and while still at the test pressure and temperature, there shall be no indication of leakage for a period of 5 minutes as checked by a pressure gage readable to 10 psi or by visual leakage of a colored test fluid. This test shall be repeated with a pressure of 3,000 psi on three units only as a qualification test (proof test). The transmitter temperature shall then be reduced to $-54^{\circ} \pm 2^{\circ}C$ by a submersion in a liquid bath at this same low temperature for a period of not less than 15 minutes. While still at this temperature, the test pressure shall be applied 5 times. After the last application of pressure, there shall be no evidence of leakage for a period of 5 minutes when checked in the same manner used for the room temperature portion of this test.

4.6.6 Response time. The time required for the transmitter output signal to come within 1 percent of the final steady state signal shall not exceed the following values for the ranges indicated when a step change in flow rate is applied. The flow rate shall be changed from approximately 2 percent of full scale to a full scale indication by the activation of a quick opening valve. The timer will be started simultaneously with the activation of the valve.

Max Range-PPH	Response time-Seconds
0-800	2.0
0-1,200	2.0
0-1,800	2.25
0-2,500	2.5
0-4,000	3.0
0-5,000	3.0

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4.6.7 Burn-in. The transmiter shall be placed in a chamber with the temperature maintained at 100°C for a minimum of 4 hours. The temperture shall then be lowered and maintained at 50°C for 20 hours with fuel flowing equivalent to half-scale indication. At least the first 8 hours of this test shall be performed during normal working hours. Observation shall be made and recorded at least once every 2 hours throughout the 8-hour period. At no time shall the transmitter output change by more than one-half the tolerance specified for the test point in Table III. Pointer oscillation or erratic movement shall be cause for rejection. The transmitter shall be returned to room temperature for a minimum of 2 hours after which it shall meet the test specified in 4.6.3. If the burn-in test is conducted other than as specified herein, the test shall be subjected to the approval of the contracting activity.

4.6.8 Position error. Position error shall be determined at 50 percent of full scale. Readings shall be taken in the normal position and with the transmitter turned 90° in any direction. The reading taken in any position shall not differ from the reading taken in the normal position by more than the value specified in Table IV.

4.6.9 Voltage and frequency variation. The applied voltage and frequency for this test shall be set at the following points:

	Voltage	Hz
Normal condition	115	400
	104.0	320
	104.0	480
	126.0	320
	126.0	480

The voltage and frequency test shall be conducted at 25 and 90 percent of full-scale flow rates. Readings shall be taken a the 115V, 400 Hz normal condition and at the conditions specified herein. The reading taken at each of the above conditions shall not differ from the reading taken at the normal condition in excess of the value shown in Table III. The transmitter shall be operated at least 1/2 hour at each of the specified conditions before any readings are taken. These tests shall be run with a master indicator and a master power supply (see 4.3.5).

4.6.10 Pressure drop. The pressur drop across the transmitter shall be determined at the flow rates specified in Table I and shall be recorded both with the transmitter energized and unenergized. The pressure drop shall be obtained by sensing the live pressure at the inlet and outlet ends of the flowmeter and obtaining the difference between the two readings. The pressure drop shall not exceed the values specified in Table I.

4.6.11 Voltage surge. A master indicator need not be used during the surge application of this test. A second harmonic receiver, or simulated receiver compatible with the transmitter will be adequate. A master power supply shall be used.

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The test shall be made at approximately 50 percent of full scale. Power shall be applied at 115 ±1V, 400 ±2 Hz for 1 hour. At the end of this operating period, the power input shall be adjusted to 115 ±1V, 320 ±2 Hz. A voltage surge shall then be applied to provide an rms peak voltage of 160 ±5V and return to 115 ±1V. The duration of the surge including rise and decay time shall be 0.09 ±0.005 second. The power input shall then be adjusted to 115 ±1V, 480 ±2 Hz and a voltage surge applied to provide an rms peak voltage of 195 ±5V and return to 115 ±1V. The duration of this surge including rise and decay time shall be 0.07 ±0.005 second. At the conclusion of this procedure, the scale error test shall be conducted at room temperature. The errors shall not exceed the room-temperature tolerances of Table III.

4.6.12 Power consumption. With 115V 400 Hz and power from the indicator to the transmitter motor applied to the transmitter, power consumption of the transmitter motor shall not exceed 20 va and power consumption of the transmitter pickoff shall not exceed 14 va.

4.6.13 Low temperature. The transmitter shall be placed in a test chamber and maintained at a temperature of -54° $\pm 2^{\circ}$C for a period of 4 hours. At the end of this period and while still at the specified low temperature, the transmitter shall be subjected to the tests specified in 4.6.3 and 4.6.4. The scale and friction errors shall not exceed the tolerances specified in Tables III and IV, respectively.

4.6.14 High temperature. The transmitter, while inoperative, shall be subjected to a temperature of 121° $\pm 2^{\circ}$C for a period of 4 hours after which it shall be tested at the specified high temperature for scale error and friction error as specified in 4.6.3 and 4.6.4. The scale and friction errors shall not exceed the tolerances specified in Tables III and IV, respectively.

4.6.15 Vibration. The vibration test shall be conducted in accordance with MIL-STD-810, method 514, equipment category (b), curve H, at room temperature with fuel flowing at approximately 50 percent of full scale. Pointer oscillation and variation shall not exceed the tolerance specified in Table IV for 10g. At the completion of this test, the transmitter shall be subjected to the tests specified in 4.6.3 and 4.6.4. Scale and friction errors shall not exceed the tolerances specified in Tables III and IV, respectively.

4.6.15.1 Vibration endurance. The transmitter shall be subjected to an endurance vibration cycling test for 40 hours in each of the three mutually perpendicular axes. The vibration spectrum shall be in accordance with MIL-STD-810, method 514, equipment category (b), curve L. The duration shall be in accordance with the figure titles "Logarmithmic Sweep". During this test, the scale error test shall be conducted at 10, 20, 40, 60, and 100 percent of full scale. Scale error shall not exceed the tolerances specified in Table IV for vibration at 20g. After completion of this test, the scale error test shall again be conducted. Scale errors shall not exceed the tolerances specified in Table III for high and low temperature. The transmitter shall be completely disassembled and examined for damage or wear that might result in failure during the design life of the instrument.

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4.6.16 Low temperature exposure. The transmitter shall be placed in a test chamber and the chamber cooled to and maintained at a temperature of -62°C for a period of 72 hours at which time the equipment shall be inspected for deterioration. The temperature of the chamber shall then be raised to -54°C and maintained for an additional 24-hour period. At the end of the 24-hour period and while still at -54°C, the transmitter shall be subjected to the scale error and friction error tests. The transmitter shall then be maintained at room temperature for a period of not less than 4 hours after which it shall be subjected to the scale error and friction error tests at room temperature. The transmitter shall be examined and there shall be no deterioration of nor damage too the transmitter as a result of the extreme temperatures exposure.

4.6.17 Temperature-altitude. The transmitter shall be placed in a test chamber and the chamber cooled to and maintained at a temperature of -54° ±2°C and at a pressure equivalent to 100,000 feet for a period of 4 hours. The transmitter shall be operated with no fuel flowing. Immediately after the 4-hour period at the specified temperature and pressure, the transmitter shall be returned to normal room conditions and the scale error, and friction error tests shall be conducted. The errors shall not exceed those specified in Tables III and IV for room-temperature conditions. This test may be combined with the test specified in 4.6.16. There shall be no damage to nor deterioration of the transmitter as a result of this test.

4.6.18 High temperature exposure. While inoperative, the transmitter shall be subjected to a temperature of 121° ±2°C for a period of 4 hours. The temperature shall then be raised to 190° ±2°C for a period of 1 hour. This cycle shall be repeated for a total of 48 hours at 121° ±2°C and 12 hours at 190° ±2°C during which fluid may be inside the transmitter but not flowing. The transmitter shall be energized and operated for the last two temperature cycles after which it shall be operated at 140° ±2°C for 4 hours at 1,000 pph with fuel flowing. The temperature of the fuel shall be 70° ±2°C. The transmitter shall then be immediately tested for scale error and friction at 121° ±2°C ambient air and the fuel at 50° ±2°C. The tolerances shall not exceed those specified in Tables III and IV. The transmitter shall be examined and there shall be no evidence of failure.

4.6.19 Pressure leakage-high temperature. The transmitter shall be subjected to a pressure leakage at high temperature test as follows: A nonshock pressure shall be applied to the inlet connection with the outlet connection bled free of air and capped. The transmitter shall be heated in a 140°C chamber for 1/2 hour and a pressure of 1,000 psi shall then be applied rapidly five times. After the last application of pressure and while still at the test pressure and temperature, there shall be no indication of leakage for 5 minutes as checked by a pressure gage readable to 10 psi or by visual leakage of a colored test fluid.

4.6.20 Humidity. The transmitter shall be subjected to a humidity test in accordance with MIL-STD-810, method 507 ater which it shall be subjected to the tests specified in 4.6.3 and 4.6.4.

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4.6.21 Fungus. The transmitter shall be subjected to a fungus test in accordance with MIL-STD-810, method 508, procedure I. at the conclusion of this test, the transmitter shall be visually examined. There shall be no deterioration or corrosion of internal components as a result of this test.

4.6.22 Salt fog. The transmitter shall be subjected to a salt fog test in accordance with MIL-STD-810, method 509, procedure I, for a period of 50 hours. The scale error and friction error tests shall be conducted before and after this test. The errors shall not exceed the low and high temperature tolerances. The transmitter shall also pass the test specified in 4.6.2.

4.6.23 Electromagnetic interference. The transmitter shall be subjected to electromagnetic conducted, radiated, and susceptibility tests in accordance with MIL-STD-462 for class ID equipment except the frequency range shall be from 0.15 to 150 MHz.

4.6.24 Shock. The shock test shall be in accordance with method 516, procedure V of MIL-STD-810. The transmiter shall be subjected to impact shocks of 25g each shock impluse having a time duration of 6 ±2 milliseconds. Three shocks shall be applied in each of the three mutually perpendicular axes.

4.6.25 Contaminated fuel operation. Test fuel containing dirt to the extent of 800 grams of foreign matter to 1,000 gallons of fuel shall be circulated through the transmitter at approximately 2,000 pph for 1 hour. Scale error and friction error tests shall then be conducted using the contaminated fuel Operation shall be satisfactory and within the room temperature tolerances. The foreign matter shall consist of not less than 68 percent Si0 and shall have a particle-size analysis as shown in Table V.

TABLE V. Contaminated Fuel Particle Size.

Foreign Particle Size (Microns)	Percent of Total Weight
0-5	39 ±2
5-10	18 ±3
10-20	16 ±3
20-40	18 ±3
over 40	9 ±3
Through a 200 mesh screen	100 percent

4.6.26 Substitute fuel. This test shall be conducted using JP-5 fuel conforming to MIL-T-5624 and at room temperatures. The scale errors shall not exceed the tolerances specified in Table III for high and low temperature.

4.6.27 Frequency cycling. The master indictor need not be used for the cycling portion of this test. A second harmonic, or simulated receiver, compatible with the transmitter will be adequate. A master power supply shall be used. The voltage shall be adjusted to 115 ±1V and the frequency cycled from 320 ±2 to 480 ±2 Hz in accordance with the following schedule with approximately half-scale flow. This constitutes one cycle for a total time for 100 seconds.

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a. Start cycle at 320 Hz, increase frequency to 480 Hz in 38 seconds lineraly with time.

b. Return frequency to 320 Hz in 2 seconds or less

c. Increase frequency to 480 Hz in 58 seonds linearly with time

d. Return frequency to 320 Hz in 2 seconds or less

The cycle shall be repeated continously for 10 hours at the rate of 36 cycles per hour. Power shall then be removed for a maximum of 2 hours and reapplied for 1 hour at 120 ±1V 320 ±2 Hz. Immediately following the 1 hour of operation at 120V 320 Hz, the transmitter shall be frequency cycled for 1 additional hour on the schedule indicated above, except that the frequency range shall be increased from 300 to 500 Hz. The transmitter shall then be subjected to the scale error test at room temperature. The scale error shall not exceed the tolerances specified in Table III for room temperature.

4.6.28 Compatibility. Compatibility tests as required to demonstrate compliance with 3.4.4, 3.4.6 through 3.4.6.3 and 4.6.3, 4.6.6, and 4.6.9 shall be conducted in accordance with the first article test procedure or the applicable contract document for the transmitters specified herein.

4.6.29 Reliability.

4.6.29.1 First article. The 11 transmitters specified in 4.4.1 shall be subjected to a reliability test in accordance with MIL-STD-781, test level G, which shall demonstrate a minimum acceptable MTBF of 1,200 hours to a 90 percent confidence level.

4.6.29.1.1 Test equipment. The test equipment and fixtures shall include provisions for transmitter operation, temperature cycling and vibration.

4.6.29.1.2 Operation. During this test, fuel at room temperture shall be flowing through the transmitter at a rate from 5 to 90 ±5 percent of full scale and return at a rate of 2 ±1 cpm.

4.6.29.1.3 Temperature cycling. Unless otherwise specified or amended by the contracting activity due to the results of the thermal analysis, the temperature of the chamber in which the transmitters are tested shall be cycled as follows:

a. Starting at room temperature, decrease temperatures to -54°C and stablize for 1 hour with transmitters unenergized.

b. Turn transmitters on, raise temperature to 95°C and operate transmitters for 3 hours.

c. Turn transmitters off and lower chamber temperature to -54°C for 1 hour.

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d. Repeat steps b and c until reliability test requirement has been met. This schedule provides six 4-hour test cycles for each 24-hour period. In each 24-hour period the operational test time is 18 hours.

4.6.29.1.4 Vibration cycling. Throughout this test, the transmitter shall be vibrated for a period of 10 minutes during each hour on ON time at any nonresonant frequency between 20 and 60 Hz and at a peak aceleration of 2.2g ±10 percent. During this test, the transmitter motor shall undergo voltage cycling from 65V 400 Hz to 126V 400 Hz linearly in time at a rate of 1 cycle every 4 ±1 minutes during operational ON time starting with 115V 400 Hz.

4.6.29.1.5 Performance check. At each 200-hour interval and at the end of this test, the transmitters shall be subjected to the scale error, friction, and response time tests. During each scale error test, if any two test points exceed twice the allowable tolerances for high and low temperature specified in Tables III and IV one failure shall be charged against this test.

4.6.29.1.6 Failure criteria. For an accept decision, each transmitter shall be operated at least 700 hours. If a transmitter fails, entries shall be made on the appropriate data logs and the failed equipment shall be removed from the test and repaired with minimum interruption of the transmitters still under test. The contracting activity shall be immediately notified of the failure and the plan for correcting the deficiency. A detailed analysis of the failure will be submitted as soon as it has been completed. No more than a total of 12 failures will be permitted during this test. Acceptance shall be based on the accept-reject criteria specified in MIL-STD-781, test plan I.

4.6.29.1.7 Reporting. Reporting shall be submitted in accordance with MIL-STD-781. The reliability test log and failure summary and analysis shall be included in the final test report.

4.6.29.2 Quality conformance. The sample transmitters specified in 4.5.2.1 shall be subjected to a reliability test in accordance with the "Reduced Testing" requirements of MIl-STD-781, test level J, and as specified under 4.6.29.1 which shall demonstrate a minimum acceptable MTBF of 1,200 hours to a 90 percent confidence level. Accept-reject criteria shall be in accordance with test plan V except the discrimination ratio shall be 1.5. When "Tightened Testing" is required by the quality activity, acceptance shall be in accordance with test plan I or III as determined by that activity. Results of this test shall be included in the test report specified in 4.6.29.1.7.

4.6.30 Longevity. Testing on 2 of the 11 reliability test samples shall be continued through a longevity test in accordance with MIL-STD-781, test plan XXVIII, until 2,500 hours have accumulated on each transmitter.

4.6.31 Maintainability demonstration. Maintainability demonstration shall be in accordance with MIL-STD-471, apendix B, test method 1. Test plan A_2 and B_1 shall be used to demonstrate a M_{ct} of 100 minutes and a $M_{max\ ct}$ of 300 minutes.

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4.7 Inspection of packaging. The inspection of the preservation packaging and interior package marking shall be in accordance with Group A and B quality conformance inspection requirements, Section 4 of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of applicable container specification and the marking requirements of MIL-STD-129.

5. Packaging

5.1 Preservation. Preservation shall be level A, C, or Industrial as specified. (see 6.2).

5.1.1. Level A.

5.1.1.1 Cleaning. Transmitter shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Transmitter shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Not applicable.

5.1.1.4 Unit packaging. Unless othewise specified by the contracting activity, each transmitter shall be packaged in quantity unit packs of one each in accordance with Method 1C1 of MIL-P-116. Overbox completed pack in PPP-B-636 container. Apply sufficient cushioning material between bag and unit container of a type, density and thickness to insure shock transmission does not exceed peak values in G's established for the transmitter when completed packs are subjected to the rough handling drop tests of MIl-P-116.

5.1.2 Level C. Each transmitter shall be clean, dry, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.1.3 Industrial. The Industrial preservation of transmitter shall be in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, C, or Industrial as specified (see 6.2).

5.2.1 Level A. Transmitter packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tate and cube consistent with the protection required.

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5.2.2 Level B. Transmiter packaged as specified in 5.1.1.1 shall be packed in shipping containers conforming to PPP-B-636, class weather-resistant, unless otherwise specified by the contracting activity. Other requirements as specified in 5.2.1 apply.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4 Industrial. The packaged transmitter shall be packed in accordance with ASTM D3951.

5.3 Marking. In addition to any other markings required by the contract or order (see 6.2) interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The TRU-31A/A transmitters covered by this specification are intended for use in measuring and transmitting a signal proportional to the total mass rate of fuel flow in pounds per hour being consumed by the aircraft engine. Accurate fuel flow measurement is essential in determining station endurance, flying time remaining and maximum engine operating efficiency as well as providing valuable engine maintenance information.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Full scale pph flow rate (appropriate transmitter type number)

c. Levels of packaging and packing required (see section 5).

6.3 First article. A first article is required for inspection and approval (see 3.1 and 4.4).

6.3.1 Test procedures. The first article test procedures shall be submitted for formal approval before the first article testing program is initiated.

6.3.2 Test samples. The first article test samples will be submitted to the contracting activity.

6.4 Subject term (key word) listing.

EFU 15A/A
Fuel, JP-5
Rate-Of-Flow, PPH
Second harmonic
Transmitter
TRU-31A/A
Turbo-Jet engine

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6.5 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with repect to the previous issue due to the extensiveness of the changes.

Custodian:
Air Force -99

Reviewer Activity:
Air Force -11

Preparing activity:
Air Force -71

Project number:
6620-F432

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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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